

WHAT ESCAPED, AND HOW TO FIND IT

Testing the plants and animals around you for engineered genes — in any country, with equipment you can buy, borrow or improvise, and no laboratory of your own.

TEN STEPS · THREE KINDS OF TEST · NONE OF THEM REQUIRES A DEGREE OR AN INSTITUTION

Japan grows no genetically engineered oilseed rape. It imports a great deal of it, and the trucks that carry it from the docks to the crushing plants leak. From 2003, citizens' groups walked those roads with protein test strips in their pockets and found feral rape carrying either the *bar* gene or CP4 EPSPS. Government scientists followed: roughly 7,500 feral *Brassica napus*, 300 *B. rapa* and 5,800 *B. juncea* seedlings screened from 143 locations, with herbicide-tolerance transgenes found at five of six major ports and along two of four sampled roadsides in the Kantō district. The strip that started it costs about what a coffee costs.

This guide is about doing that work where you live. It assumes no laboratory, no institutional affiliation, no science degree and no budget beyond a few hundred units of your currency. It assumes you can walk, photograph, take notes, put a leaf in a paper envelope and post it. Ten steps follow, in the order you should do them. The numbers match the badges on the map overleaf.

FOUR DIFFERENT THINGS PEOPLE MEAN BY "ESCAPE"

Decide at the start which one you are looking for, because they need different evidence and they carry different weight.

A NOTE ON THE WORDS USED HERE

Escape, feral, volunteer, contamination. These are the terms used in permits, dossiers and journals, and they appear in this guide so that you can search the same words the authorities do. They are not neutral, and this guide does not endorse them. *Escape* assumes the organism had somewhere it was supposed to stay. *Feral* defines a lineage by whether it is still serving somebody. *Contamination* treats one genome as clean and another as spoiled. Each of them grades living things by their usefulness to a producer.

VOLUNTEERS. Last season's engineered crop coming up in this season's field, or in the ditch beside it. Same genotype as the crop that was planted. Easy to find, weak as a finding, because nobody disputes that volunteers happen.

FERALS. Populations sustaining themselves outside cultivation — road verge, railway ballast, riverbank, port apron, mill yard. Oilseed rape is the standard case because the seed is small, round and spills. Switzerland permits no GM cultivation and still found feral GT73 rape along the rail lines running from the Italian and French borders to the crushing plants in Ticino and around Basel.

HYBRIDS. The transgene has crossed into a different population: a neighbour's conventional crop, a landrace, a wild or weedy relative. This is the finding that matters most and the hardest to establish, because you have to show both that the transgene is there and that the plant carrying it is not simply a spilled crop seed that germinated.

CONTAMINATED LOTS. The transgene arrived in a sack. It never crossed a field boundary; somebody sold it to you, or donated it, or fed it to your animals. Seed, grain, birdseed and relief seed are all routes, and unapproved events turn up in them regularly.

THREE SIGNALS, THREE PRICE BRACKETS

There are only three properties of an engineered organism that anyone can measure, and each has its own instrument, cost and blind spot.

PROTEIN. The transgene makes something — CP4 EPSPS for glyphosate tolerance, PAT or BAR for glufosinate, a Cry or Vip protein for insect toxicity. A lateral-flow strip detects that protein in a crushed leaf or seed in five to ten minutes with no equipment at all. Roughly the cost of a bus ticket per test, in boxes of 25 or 100.

DNA. The inserted sequence itself, found by PCR. This is the evidence that survives argument, and it is the one step where you will need somebody else's machine. Days to weeks; tens to low hundreds of units per sample if you pay a commercial laboratory, free if a teaching lab runs it as a class.

PHENOTYPE. What the trait actually does. Grow the seed out and spray it with the herbicide the trait confers tolerance to. Costs nothing but a season, works only for herbicide tolerance, and demonstrates the trait as an inherited property rather than a protein in one leaf.

THE THREE STRINGS THAT DO MOST OF THE WORK

P-35S. The 35S promoter of cauliflower mosaic virus, used to switch inserted genes on.

T-NOS. The nopaline synthase terminator from *Agrobacterium tumefaciens*, used to switch them off.

Surveys of commercialised events put P-35S in about 66 per cent and T-NOS in about 53 per cent, with one or the other in roughly 81 per cent. Screening for those two catches most of what is out there and misses the rest by design.

THE UNIQUE IDENTIFIER. A short code such as MON-00810-6, assigned to one engineered event and used to follow it through every database on earth. A strip gives you a protein, a screening PCR an element, and an event-specific PCR this.

AND ONE THING YOU PROBABLY CANNOT DETECT

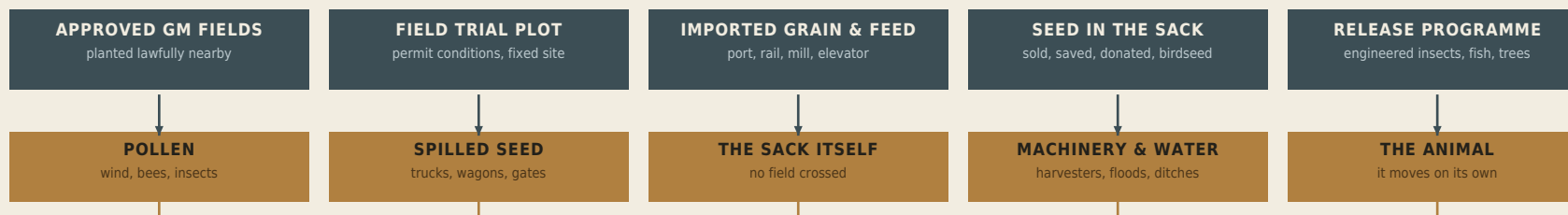
Gene editing that only cuts and repairs — the class regulators call SDN-1 — leaves no foreign DNA and often produces no novel protein. There is nothing for a strip to bind and no 35S promoter to amplify. Detection is not impossible: a quantitative PCR method for sulfonylurea-tolerant SU Canola, the first commercialised genome-edited crop, was published in *Foods* in 2020, made openly available, and validated by the Environment Agency Austria. But it required knowing the exact edit, and it took a funded laboratory to build.

If the organism near you was made this way and the developer has not disclosed the sequence, you will not find it with a dipstick and a stamp. Write that limitation

THE GROUND, THE ROUTES, AND WHAT EACH TEST CAN SEE

PAGE 2 · THE MAP

WHERE IT COMES FROM — engineered material is either planted here or moved through here

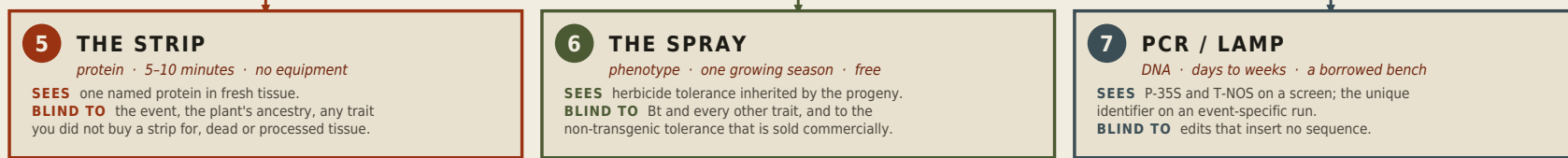


WHAT NOW CARRIES IT — the populations you sample. rust boxes are where a crossing cannot be undone



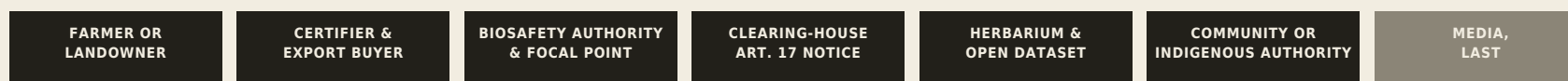
3 SAMPLE ALONG THE SPILL LINES AND THE DOWNWIND EDGES · GPS, PHOTOGRAPH, SPLIT AND DRY EVERY SAMPLE 4

WHAT EACH TEST CAN SEE — and what it is blind to



GENE-EDITED, NO INSERTED DNA (SDN-1): NOTHING FOR A STRIP TO BIND AND NO 35S TO AMPLIFY. NO TEST ON THIS PAGE SEES IT — WRITE THAT INTO THE REPORT.

WHERE THE RESULT GOES — fastest correspondents first



How to read this. Sources sit across the top: engineered material enters your area because somebody planted it or because somebody moved it through. The five routes beneath are the only ways it leaves a field or a container. The boxes below are the populations that now carry it and the ones you will sample; the two in rust, landraces and wild relatives, are where a crossing cannot be undone. The three lanes are every test available to you, with what each is blind to written into it, and the black bar beneath them is the category no test of yours reaches. The bottom row is where a finding goes, fastest correspondents first. The panel on the right is how a finding gets destroyed, usually by the person who made it. The numbered badges are the ten steps. Start at the badge nearest your own situation rather than at step one.

WHAT DEFEATS A RESULT

- 10 CaMV FALSE POSITIVE**
Cauliflower mosaic virus carries the 35S promoter. An infected non-GM brassica screens positive. Exclude the virus before you speak.
- NO CONTROLS**
A strip with no known positive and no known negative beside it is an anecdote, and gets treated as one.
- NO GPS, NO VOUCHER**
A sample nobody can relocate or re-examine cannot be defended and cannot be replicated.
- "ONLY A VOLUNTEER"**
Answer with species identity, distance from cultivation, whether it set seed in place, and a grow-out of the progeny.
- NEGATIVE IS NOT ABSENT**
Forty clean plants bounds the rate loosely and proves nothing. $n = \log(1-C) \div \log(1-p)$ gives 3,000 units for 0.1% at 95%.
- ONE PERSON, ONE COPY**
These files move over years and volunteers do not. Two people hold everything, in two buildings, from the first week.

WHAT THE KIT ACTUALLY IS

- 25 or 100 lateral-flow strips for the two most likely proteins — the only real expense in this guide
- The tubes and pestles that come in the box
- A box of nitrile gloves, changed between plants
- Paper coin or seed envelopes, never plastic
- A kilo of silica gel and a sealable jar, both reusable indefinitely
- A pencil, a permanent marker and a notebook that survives rain
- A phone, for coordinates and photographs
- A coin or folding rule, in every photograph, for scale
- Postage, and the receiving laboratory's paperwork obtained first

into your report rather than leaving it out. The gap is a finding of its own, and it is the reason a growing number of countries have written these organisms out of their biosafety law entirely.

1 DECIDE WHICH ORGANISM AND WHICH TRAIT, BEFORE YOU SPEND ANYTHING

Every test on the market is specific. A CP4 EPSPS strip tells you nothing about Bt maize; a Cry1Ab strip tells you nothing about herbicide-tolerant rape. Buying before you know what you are looking for is the commonest way a first attempt is wasted.

Answer three questions in writing before you order anything.

- **What is approved or under trial within about fifty kilometres?** Search the Biosafety Clearing-House at bch.cbd.int by country, then your national biosafety register, which usually sits under the agriculture or environment ministry. The ISAAA GM Approval Database is a serviceable cross-check on which events are cleared where.
- **Which of those crops has a wild or weedy relative growing here?** Gene flow needs a partner. A local botanist, a herbarium curator or a university botany department will answer this for free, and answering it correctly is most of the work.
- **What passes through that is not grown here?** Ports, crushing plants, feed mills, grain elevators, rail sidings and highway routes carry imported seed through places that have never planted it. Japan and Switzerland both found their ferals this way.

CROP	CAN CROSS WITH	USUAL PROTEIN
Maize	Teosinte; other maize	Cry1Ab/Ac, Cry1F, Cry3Bb, CP4 EPSPS, PAT
Oilseed rape / canola	B. rapa, B. juncea, wild mustard, charlock	CP4 EPSPS, PAT/BAR
Soybean	Glycine soja (East Asia only)	CP4 EPSPS, PAT
Cotton	Wild Gossypium (Mexico, Caribbean, Australia)	Cry1Ac, Cry2Ab, Vip3A, CP4 EPSPS
Rice	Weedy / red rice, O. rufipogon	PAT/BAR, Cry proteins
Sugar beet	Sea beet, B. vulgaris maritima	CP4 EPSPS
Alfalfa / lucerne	Feral alfalfa	CP4 EPSPS
Squash, papaya, brinjal	Local cultivars and wild forms	Coat proteins, Cry1Ac

2 WALK THE GROUND WHERE SEED IS DROPPED, NOT THE GROUND THAT IS CONVENIENT

Contamination is not spread evenly across a landscape, so sampling evenly wastes money. Two processes put engineered material outside a field: seed falls off vehicles, and pollen blows or is carried. Sample where each of those happens.

SPILL LINES. The last few hundred metres of road before a crushing plant, feed mill, elevator or port gate. Level crossings, switch yards, weighbridges and unloading points. Any verge on a route harvest trucks use. Look at the base of the kerb and in the gravel, where seed lodges.

POLLEN EDGES. The downwind margin of a conventional field within pollen distance of an engineered one, and the ditch banks, hedge bottoms and headlands between them. Isolation distances written into permits give you a number to compare your map against.

RELATIVES. Wild and weedy populations of the crop's relatives, prioritising centres of origin and diversity, where a single crossing is permanent and matters most.

SACKS. Saved seed, market seed, relief and aid seed, birdseed, and animal feed. This is where events that were never approved anywhere near you tend to appear.

HIVES. Honey and bee-collected pollen integrate a whole foraging radius into one sample. The Bavarian beekeeper Karl Heinz Bablok found MON810 maize DNA and Bt protein in pollen from hives sited within about 500 metres of a state trial plot; the case ran to the Court of Justice of the European Union in 2011.

3 WORK OUT HOW MANY SAMPLES YOU ACTUALLY NEED

The arithmetic here is short and worth knowing, because it decides whether your negative result means anything. To have confidence C of finding at least one positive when the true rate is p, the number of units you must test is:

n = log(1 - C) ÷ log(1 - p)

For 95 per cent confidence at a rate of 0.1 per cent, that gives 2,995, which the seed trade rounds to 3,000 seeds. For 1 per cent it gives 299. The International Seed Testing Association distributes a free spreadsheet, Seedcalc, that does this and the composite-sample versions of it.

IN OTHER WORDS. Rare things hide. If one seed in a thousand carries the gene, a hundred seeds will almost certainly miss it, and catching it would be luck. You need around three thousand before finding one becomes the likely outcome instead of the lucky one. If one seed in a hundred carries it, three hundred will do.

You are almost certainly not going to test 3,000 seeds, and you do not have to. Understand what the number is telling you. Finding one positive anywhere is a finding, whatever your sample size. Finding nothing in forty plants tells you the rate is probably not enormous, and nothing more than that. Report it that way. A negative described as proof of absence is the single fastest way to lose an argument you were winning.

COMPOSITE OR INDIVIDUAL. Grinding a hundred leaf discs together and running one strip is cheap and answers "is it here at all". It will not tell you which plant. Test individually where you may need to go back — for anything you suspect is a hybrid, and for anything on a wild relative.

4 TAKE THE SAMPLE SO THAT IT SURVIVES BEING CHALLENGED

FOR STRIPS, USE FRESH TISSUE. The youngest fully expanded leaf, from a living plant, tested the same day. Bt protein levels fall sharply in old and senescing tissue, so a strip run on a brown leaf can read negative on a plant that carries the gene.

FOR DNA, DRY IT FAST. Put the leaf in a small paper envelope, put the envelope in a sealed jar of silica gel, and leave it for a day or two. Properly dried leaf holds usable DNA at room temperature for years. Sealing fresh leaf in a plastic bag rots it, and the DNA goes with it. Where you can get them, DNA preservation cards let you press sap onto treated paper and post it as a letter.

LABEL EVERYTHING AT THE MOMENT OF COLLECTION. A sample number, the date and time, coordinates to five or six decimal places from any phone, a photograph of the plant standing where it grew with something for scale, a second photograph showing the surroundings, the collector's name and the name of a witness.

SPLIT EVERY SAMPLE. Halve it, store the halves in two buildings held by two people, and sign and date the log. A countersignature from somebody with no stake in the outcome costs nothing and closes an argument that would otherwise take a year.

ASK BEFORE YOU POST IT ACROSS A BORDER. Plant material generally needs a phytosanitary import permit, and soil is prohibited almost everywhere. Ask the receiving laboratory what paperwork it needs before you send anything, and never post soil.

5 THE STRIP: PROTEIN, IN TEN MINUTES, ANYWHERE

A lateral-flow strip is a piece of nitrocellulose carrying an antibody raised against one transgenic protein. Crush a leaf disc about a centimetre across, or a single seed, in the tube of buffer the kit supplies. Stand the strip in it. Wait five to ten minutes and read it.

Two lines is a positive. One line — the control line alone — is a negative. No control line means the test failed, and you throw it away and run another. Published sensitivities are around 0.1 per cent in bulk grain, and considerably better on a single leaf, where you are asking whether one plant expresses the protein at all.

WHERE TO BUY. EnviroLogix (QuickStix), Agdia (ImmunoStrip), Romer Labs (AgraStrip), Eurofins Technologies and Gold Standard Diagnostics all sell them, as do manufacturers in India and China at lower prices. Kits come as 25 or 100 strips with tubes and pestles. They are not controlled items in most jurisdictions, but check import rules for your own before ordering.

RUN CONTROLS OR DO NOT BOTHER. Every batch needs a known positive and a known negative alongside it. A kernel of the commercial GM variety serves as the positive; a certified organic seed of the same species serves as the negative. Without both, a result is an anecdote.

What a strip will not do:

- Identify the event, so it cannot tell you whose organism it is.
- Distinguish a crop volunteer from a hybrid. It sees the protein, not the plant's ancestry.
- Detect a trait you did not buy a strip for, including anything gene-edited.
- Work reliably on dead, dried, cooked or processed material.

6 THE SPRAY: PHENOTYPE, FOR THE PRICE OF A SEASON

This is the only test in the guide that costs nothing, and it is the one most people skip. It works for herbicide tolerance only.

- Collect seed from the plants in question and note which plant each lot came from.
- Sow fifty to a hundred seeds in trays, with the same number of a known susceptible variety beside them as a control.
- At the two- to four-leaf stage, apply the herbicide the trait confers tolerance to, at the rate printed on the label for field use.
- Susceptible seedlings yellow and collapse within seven to fourteen days. Tolerant ones carry on.

The reason to do it is that it tests the progeny. A strip tells you that one leaf on one plant contained a protein; a grow-out tells you the trait was inherited, which is what gene flow means. It is also how the Japanese roadside surveys confirmed their strip results, alongside immunochemical and DNA methods. If a batch survives two different herbicides, you have found stacked tolerance, which is a great deal more interesting than a single volunteer.

CAUTIONS. Non-transgenic herbicide tolerance exists and is sold commercially — Clearfield lines are the obvious case — so survival alone does not prove engineering. You need the right herbicide at the right rate. You are handling a pesticide: gloves, eye protection, no application where drift reaches anything you care about, lawful disposal, and in some countries a licence to buy or apply it at all.

7 THE DNA: THE RESULT THAT SURVIVES ARGUMENT

Two tiers, and you generally want both. **Screening** runs PCR for P-35S and T-NOS plus a plant housekeeping gene as an extraction control, and tells you that an engineered element is present. **Event-specific** PCR amplifies across the junction between the inserted DNA and the host

genome, and gives you the unique identifier.

Places that will run it, roughly in order of what they cost you:

- **A school, community college or university teaching laboratory.** Bio-Rad's GMO Investigator kit is a standard classroom exercise: PCR for the 35S promoter and the NOS terminator, with a photosystem II chloroplast gene as the extraction control. A biology teacher can run your samples as a practical, and students take the work seriously.
- **A community biology laboratory or hackerspace,** where one exists within reach.
- **An agricultural extension service, seed testing station, or the national plant protection laboratory.** Many run GMO screening already for the seed trade.
- **A researcher.** Write to the botany, agronomy or ecology department at the nearest university. Name the species, the site and what you have already found. People answer this letter more often than you would expect.
- **A commercial food-testing laboratory.** Eurofins, SGS, Mérieux NutriSciences and their national equivalents sell GMO screening as a routine service. Get a quote first, and tell them it is an environmental leaf sample rather than grain, because the extraction differs.
- **Your organic certifier.** Certification bodies test for contamination as part of their own business and lose money when it appears, which makes them fast correspondents.

ISOTHERMAL AMPLIFICATION. LAMP runs at a constant 60–65 °C for forty-five minutes to an hour and can be read as a colour change by eye, which removes the thermocycler. Primer sets targeting P-35S, NOS, *pat* and *cp4-epsps* are published. It does not remove the reagents, the enzyme, the cold chain or the need for clean technique, so treat it as something to ask a friendly laboratory about rather than a kitchen procedure.

FALSE POSITIVES AND FALSE NEGATIVES — READ THIS BEFORE YOU PUBLISH ANYTHING

THE 35S TRAP. Cauliflower mosaic virus is a real and common virus of brassicas, and the 35S promoter is its own. A non-engineered cabbage, turnip or watercress infected with it will screen positive for P-35S. Any P-35S positive on a brassica has to be followed by a CaMV-specific test or an event-specific PCR before you say a word in public. Multiplexed assays that separate the two exist and are published. Announcing a screening positive on a wild mustard without excluding the virus is how a campaign loses its credibility in one afternoon.

STRIP NEGATIVE, DNA POSITIVE. Protein expression varies with tissue, plant age, temperature and stress, and in a hybrid the transgene may be present in half the offspring or silenced altogether. A negative strip on a plant you have good reason to suspect is a reason to send tissue for PCR, not a reason to stop.

YOUR OWN HANDS. Do not grind a known GM kernel on the same board, with the same pestle, in the same session as your field samples. Change gloves between plants. Run a blank — buffer and no tissue — through the whole procedure and check that it comes back clean.

8 TURN A RESULT INTO A RECORD SOMEBODY ELSE CAN CHECK

A finding nobody can re-examine is worth very little, and it costs nothing to make one that can be. Each entry needs the date, coordinates, a photograph of the plant in place, the species and who identified it, the method with kit name and lot number, the controls and what they gave, the raw result, and a list of what you did not test for.

PRESS A VOUCHER. Flatten the plant under weight, dry it, mount it on card with species, place, date and collector, and lodge it with a national or university herbarium. Free, permanent, and anyone can re-examine the specimen in fifty years.

PUBLISH THE RAW TABLE. One row per sample, negatives included. Data with the failures removed reads as advocacy and gets treated as advocacy.

9 SEND IT TO THE PEOPLE WHO MOVE FASTEST, IN THAT ORDER

- **The farmer or landowner, first.** They carry the consequence, and finding out from a newspaper turns a possible ally into a permanent opponent.
- **Organic certifiers and export buyers.** They act on commercial grounds and commercial timescales, which are far shorter than a ministry's.
- **The national biosafety authority and the Cartagena focal point,** in writing, citing any permit file number the material relates to. If that approval carried monitoring conditions, ask for those reports by name and year; their absence is a breach of the approval itself.
- **The Biosafety Clearing-House.** Parties are obliged to notify unintentional transboundary movements of living modified organisms. Ask in writing whether yours was notified.
- **A university research group** with a validated method, who can take it further than you can.

- **Indigenous and community authorities** where the land is theirs; their institutions decide what happens with a finding on their ground. **Media last,** once you hold the confirmation.

10 WHAT THEY SAY BACK, AND WHAT ANSWERS IT

"YOUR METHOD IS NOT VALIDATED." Fair against a strip alone. Answer with event-specific confirmation from an accredited laboratory, or ask them in writing to run their own. The unanswered request is useful later.

"THAT IS A VOLUNTEER, NOT GENE FLOW." Answer with botany: species identity, distance from cultivation of that crop, the size and age of the stand, whether it has set seed in place. Or with a grow-out, which settles inheritance.

"IT IS BELOW THE LABELLING THRESHOLD." Thresholds govern labelling of a product for sale. They say nothing about a living population outside any permit.

"THE EVENT IS APPROVED." Approval to cultivate under conditions is not approval to grow in a river bed or a road verge. Ask which permit condition covers the place you found it.

SILENCE. Number your letters, ask one closed question in each, copy a second body, keep the receipts. Three unanswered numbered letters is a fact about the authority, and reads that way to a committee, an ombudsman or a court.

ORGANISMS THAT ARE NOT PLANTS

INSECTS. An ovitrap is a dark cup of water with a paper lining that mosquitoes lay on; reared larvae can be counted and kept. Engineered lines usually carry a fluorescent marker so their developers can sort them — ask which marker and which wavelength, and expect the answer in the permit file. A 2019 study reported transgenic sequence persisting in wild *Aedes aegypti* at Jaboticabal, Brazil after an Oxitec release; the company disputes it and the journal added an expression of concern.

FISH. Fluorescent ornamental zebrafish escaped farms in south-eastern Brazil and were caught in all five sampled headwater creeks of the upper Paraíba do Sul basin, where their sale is banned. You can see them. Photograph, locate, preserve a specimen; hybrids need PCR, for which a published assay exists.

MICROBES. Engineered soil bacteria and fungi are beyond any citizen method. Turn the question back on the file: who monitors, on what schedule, measuring what. In a great many approved dossiers, nobody does.

EIGHT WAYS PEOPLE WASTE A YEAR OF THIS

1. Buying strips for a trait not present in the area.
2. Reporting a P-35S positive on a brassica without excluding cauliflower mosaic virus.
3. Running no positive and no negative control.
4. Sampling on a tidy grid rather than the spill lines and downwind edges.
5. Reporting a negative as proof of absence.
6. Sealing fresh leaf in plastic, or posting soil.
7. One person holding the samples and the only copy of the data.
8. Sampling private land unasked, so the argument becomes how you obtained it.

YOUR FIRST WEEKEND

SATURDAY. Search bch.cbd.int and the national register for what is approved, imported or under trial nearby, and write down the events, traits and proteins. Then walk the route from the nearest port, mill or elevator, photographing every stand of the crop or its relatives on the verge, with coordinates. Collect nothing yet.

SUNDAY. Ask a local botanist which relatives grow there, and ask the nearest biology teacher whether they hold a PCR kit and would run a class practical. Order the two strips matching the likeliest two proteins. Put the photographs and coordinates in a shared folder; give two other people access.

WHERE TO LOOK, AND WHO ANSWERS STRANGERS

Documents: the Biosafety Clearing-House at bch.cbd.int; your national biosafety register; the ISAAA GM Approval Database; the EU GMO and EFSA application registers; APHIS and regulations.gov in the United States. **Methods:** the EU Reference Laboratory for GM Food and Feed publishes validated event-specific PCR methods free. **Tools:** the ISTA Seedcalc spreadsheet.

People who answer a stranger: Testbiotech and GMWatch on dossiers and detection; Third World Network's Biosafety Information Centre for the global South; the African Centre for Biodiversity; the Environmental Law Alliance Worldwide for legal help in your own country; your national organic certifier association; and your country's Cartagena focal point.

IF YOU DO ONLY ONE THING THIS MONTH · Walk the last kilometre of road before the nearest crushing plant, feed mill or port gate, in the season the crop flowers, and photograph every stand of it or its relatives on the verge, with coordinates. Start there because spilled seed collects on that stretch, so one afternoon on it finds more than a week spread evenly across the district — and it produces the map everything else in this guide runs from.